

altairsim — examples

2026-08-22 · 5d2887e

Machines that boot. Each directory here is self-contained: a `.toml` that describes the machine, the media it needs lying beside it, and a note saying what you will see. Copy any one of them anywhere and it still runs — a path inside a machine file resolves against **that file**, not against the directory you launched from.

In the release

These eight are what the distribution zip carries — the machines `docs/manual/quick-start.md` names. Each is a `DIR` line in `docs/package.map`, `tools/build-package.sh` assembles them, and `acceptance-examples` boots every one out of a scratch directory with no repository in sight — because "does it work here" and "does it work where we hand it to people" turned out to be different questions.

```
altairsim examples/cpm/cpm22-buffered.toml    # CP/M 2.2b on an 8" floppy
altairsim examples/cpm/cpm22-terminal.toml    # ...the same, console in a built-in VT100 window
altairsim examples/basic/basic4k.toml         # Altair 4K BASIC, off a 1975 cassette
altairsim examples/diskbasic/diskbasic.toml    # Altair Disk Extended BASIC 4.1 on an 8" floppy
altairsim examples/hdsk/hdsk.toml             # CP/M off an 88-HDSK hard disk
altairsim examples/acr/mitstapes.toml         # CP/M + WRTAPE, writing MITS cassettes to an 88-ACR
altairsim examples/printing/printer.toml      # an 88-C700 printer, wired to a real host printer
altairsim examples/debugger/debugger.toml     # a bench for learning the symbolic debugger
```

	What it is
<code>cpm/</code>	Mike Douglas's track-buffered CP/M 2.2b v2.3 , 56K, booted by the DBL PROM from an 8" floppy. <code>A></code> in one command. <code>cpm22-terminal.toml</code> boots the same disk with its console in a built-in VT100 window the simulator draws itself — no telnet client, no external emulator.
<code>basic/</code>	Altair 4K BASIC 3.1 (<code>basic4k.toml</code>) read off a period <code>.tap</code> by the bootstrap MITS shipped, unmodified: <code>MEMORY SIZE?</code> . Beside it, 8080 BASIC VER 1.0 (<code>basic1.toml</code>) — the first Altair BASIC, 1975 — booted the raw two-step way its primitive bootstrap forced: <code>RUN 1800</code> , <code>^E</code> , <code>RUN 0</code> → <code>MEMSIZ?</code> . Both also come as <code>.wav</code> audio cassettes for the 88-ACR (<code>basic4k-wav.toml</code> , <code>basic1-wav.toml</code>).
<code>diskbasic/</code>	Altair Disk Extended BASIC 4.1 (MITS, 1977) on an 8" floppy behind an 88-DCDD, booted by the DBL PROM. Unlike the cassette BASIC, this one has a filesystem — <code>SAVE</code> by name, a directory, <code>DSKINI</code> . <code>MEMORY SIZE?</code>
<code>hdsk/</code>	CP/M 2.2 booting off an Altair 88-HDSK "Datakeeper" hard disk — an outboard controller moving whole sectors over an 88-4PIO — loaded by the HDBL PROM. <code>A0></code>
<code>acr/</code>	Mike Douglas's MITS Tapes CP/M disk and WRTAPE , the utility that writes any of the MITS distribution BASICs back out through an 88-ACR — a bootable audio cassette you can then load on <code>basic4k</code> / <code>basic8k</code> / <code>ps2</code> , or on real hardware.
<code>printing/</code>	An 88-C700 line printer , and a banner program that prints through it. The README sets up a real printer on your host — a network printer over <code>socket:</code> , or a CUPS queue over <code>printer:</code> — and a page comes out. Per-OS host setup (macOS, Linux; Windows pending).
<code>debugger/</code>	A 46-byte program with its symbols and a guided walk through the monitor's debugger: <code>SYMBOLS LOAD</code> , symbolic <code>DISASM</code> , single-step, break on a label, run.
<code>ai-mcp/</code>	A working directory for an AI assistant driving altairsim over MCP : a CP/M machine and a tiny <code>HELLO.ASM</code> with one deliberate bug the assistant assembles, runs, single-steps to find, and fixes — all through the simulator's MCP tools. See <code>DRIVING-WITH-AI.md</code> .

More machines — the `altairsim-machines` companion

There are more folders under `examples/` than the nine above: Cromemco boards, the SD Systems and Tarbell and iCOM disk systems, a Sol-20, the dual-card storage boards, and others. They are real, tested machines — each keeps its acceptance test — that simply do not travel in the core zip. They are published on their own through `altairsim-machines`, a companion distribution that mirrors them out of this tree. Browse the folders here to see what is available; each carries its own README saying what it is and what to type.